Empirical Analysis Section

6 Empirical Analysis

In this section, we present empirical evidence from the Wine Reviews dataset to validate the proposed FT+PPI framework. After introducing the dataset, estimand, label accounting, and LLM fine-tuning protocol in Section 6.1, we conduct four empirical analyses. First, in Section 6.2, we assess the goodness-of-fit of the LoRA-Var scaling law on an independent pilot set. Second, in Section 6.3, we test whether the allocation rule implied by this pilot scaling law is close to a full-grid empirical allocation benchmark on the target population. Third, in Section 6.4, we evaluate a target-only ramp-up procedure that estimates a low-regret allocation without training on the full grid. Fourth, in Section 6.5, we compare the resulting FT+PPI estimators with natural benchmarks, focusing on estimation accuracy, variance reduction, and sample savings. Section 6.6 then discusses how the single-split allocation study can be combined with CrossPPI.

The empirical design keeps the pilot analysis separate from the target-population inference experiments. An independent pilot set is used to study the residual-variance scaling law and is never reused when evaluating allocation, ramp-up, or final estimation. All target-budget experiments draw fresh labeled datasets from a disjoint target population. This split lets the scaling-law evidence play the same role as a design-stage calibration, while the reported inference results remain tied to fresh labeled budgets from the target population.

6.1 Empirical Setup

Textual product reviews are a common source of information for assessing perceived quality in many markets. Such reviews encode rich but noisy signals about latent product attributes that are difficult to measure directly. In the Wine Reviews setting, the text describes aroma, flavor, balance, and finish, while the expert rating summarizes these signals into a numerical assessment.

This setting is especially relevant when high-quality evaluations are costly. Each additional labeled review consumes expert attention, and improving statistical efficiency directly reduces the number of labels needed to reach a target level of precision. Thus, variance reduction is not only a technical objective; it determines how reliably one can form aggregate assessments of product quality under a fixed evaluation budget.

Dataset and Estimand. The data come from the Wine Reviews dataset, which contains reviews written by professional wine critics at *Wine Enthusiast* magazine and publicly available on Kaggle (Thoutt 2017). This dataset has also been used in prior work on prediction-powered inference and surrogate outcomes (Ji et al. 2025). Each observation contains a free-form review text and an expert rating from 80 to 100 points. We use only the review text as the model input and the expert rating as the outcome.

After dropping observations with missing review text or rating and deduplicating by review description, the cleaned dataset contains 119,955 unique reviews. Due to the computational cost of repeated LLM fine-tuning, we randomly sample 25,000 reviews from this cleaned dataset and use them as the experimental finite population

$$\mathcal{P}_0 = \{(X_i, Y_i) : i = 1, \ldots, N_0\}, \qquad N_0 = 25{,}000.$$

The raw rating variance in the cleaned data is approximately 9.57. This number is the residual variance of a constant predictor and is the baseline for judging whether fine-tuned LLM surrogates extract useful information from text. Table 1 reports two representative examples from the Wine Reviews dataset.

Table 1: Examples from the Wine Reviews dataset

Review (X)	This is a walk backward after the impressive 2012. Almost impenetrably black, the flavors converge around espresso and bitter chocolate, yet the tannins have a green edge. The wine simply feels flat in the mouth with no life to it.
Rating (Y)	85
Review (X)	This is one of the great Rieslings from the Wachau, a wonderful panoply of ripe, tropical fruit, pierced with flint, spice and minerality. It is rich and opulent, while never losing sight of the core tautness of a fine Riesling.
Rating (Y)	95

Pilot and target populations. To separate scaling-law estimation from downstream inference, we partition the experimental population once into two disjoint sets:

$$\mathcal{P}_0 = \mathcal{H}_{\text{scale}} \,\dot\cup\, \mathcal{P}_{\text{target}}.$$

The pilot set has size $|\mathcal{H}_{\text{scale}}| = 10{,}000$, and the target population has size $|\mathcal{P}_{\text{target}}| = 15{,}000$. The set $\mathcal{H}_{\text{scale}}$ is used only in Section 6.2 to assess whether LoRA-Var residual variance follows a stable power law. It is not used to choose or evaluate any target-budget estimator. All subsequent experiments draw labels only from $\mathcal{P}_{\text{target}}$, and the finite-population mean of $\mathcal{P}_{\text{target}}$ is the estimand for Sections 6.3–6.5.

PPI Estimator Construction. For the target-population experiments, each replication begins by drawing a fresh small labeled dataset

$$\mathcal{N}_r^{(B)} \subset \mathcal{P}_{\text{target}}, \qquad |\mathcal{N}_r^{(B)}| = B,$$

by simple random sampling without replacement. The remaining reviews

$$\mathcal{M}_r^{(B)} = \mathcal{P}_{\text{target}} \setminus \mathcal{N}_r^{(B)}$$

form the large unlabeled text dataset. As in the main framework, the estimator may use the texts X_i for $i \in \mathcal{M}_r^{(B)}$, but the labels Y_i for these reviews are never used for training, model selection, allocation selection, scaling-law fitting, correction, or estimator construction. This convention makes the target-population experiments finite-population inference problems rather than prediction benchmarks.

Within the labeled budget, we first split

$$\mathcal{N}_r^{(B)} = \mathcal{V}_r^{(B)} \,\dot\cup\, \mathcal{N}_{r,\text{eff}}^{(B)}.$$

The validation subset $\mathcal{V}_r^{(B)}$ is used for checkpoint selection and, when applicable, for target-only ramp-up and allocation selection. The remaining labels $\mathcal{N}_{r,\text{eff}}^{(B)} = \mathcal{N}_r^{(B)} \setminus \mathcal{V}_r^{(B)}$ are the only labels available for the final estimator. For a candidate allocation s, we further split

$$\mathcal{N}_{r,s}^{\text{FT}} \subset \mathcal{N}_{r,\text{eff}}^{(B)}, \qquad |\mathcal{N}_{r,s}^{\text{FT}}| = s, \qquad \mathcal{N}_{r,s}^{\text{PPI}} = \mathcal{N}_{r,\text{eff}}^{(B)} \setminus \mathcal{N}_{r,s}^{\text{FT}}.$$

The set $\mathcal{N}_{r,s}^{\text{FT}}$ is used for fine-tuning, and $\mathcal{N}_{r,s}^{\text{PPI}}$ is used for PPI correction. Thus, validation, fine-tuning, and correction labels are disjoint within each replication and budget. Table 2 summarizes the nested decomposition and role of each split.

Table 2: Nested data decomposition in the empirical protocol

Set	Composition / size	Role
$\mathcal{P}_0$	$\mathcal{H}_{\text{scale}} \,\dot\cup\, \mathcal{P}_{\text{target}}$, $\lvert\mathcal{P}_0\rvert = 25{,}000$	Experimental finite population.
$\mathcal{H}_{\text{scale}}$	$\lvert\mathcal{H}_{\text{scale}}\rvert = 10{,}000$	Used only for independent scaling-law validation in Section 6.2.
$\mathcal{P}_{\text{target}}$	$\mathcal{P}_0 \setminus \mathcal{H}_{\text{scale}}$, $\lvert\mathcal{P}_{\text{target}}\rvert = 15{,}000$	Target finite population for allocation, ramp-up, and inference experiments.
$\mathcal{N}_r^{(B)}$	$\mathcal{N}_r^{(B)} \subset \mathcal{P}_{\text{target}}$, $\lvert\mathcal{N}_r^{(B)}\rvert = B$	Small labeled dataset drawn in replication r.
$\mathcal{M}_r^{(B)}$	$\mathcal{P}_{\text{target}} \setminus \mathcal{N}_r^{(B)}$	Large unlabeled text dataset; labels are never used.
$\mathcal{V}_r^{(B)}$	$\mathcal{V}_r^{(B)} \subset \mathcal{N}_r^{(B)}$ (20% of B in allocation experiments)	Validation for checkpoint selection and allocation/ramp-up decisions.
$\mathcal{N}_{r,\text{eff}}^{(B)}$	$\mathcal{N}_r^{(B)} \setminus \mathcal{V}_r^{(B)}$	Effective labels available to the final FT+PPI estimator.
$\mathcal{N}_{r,s}^{\text{FT}}$	$\mathcal{N}_{r,s}^{\text{FT}} \subset \mathcal{N}_{r,\text{eff}}^{(B)}$, $\lvert\mathcal{N}_{r,s}^{\text{FT}}\rvert = s$	Labels used to fine-tune the surrogate.
$\mathcal{N}_{r,s}^{\text{PPI}}$	$\mathcal{N}_{r,\text{eff}}^{(B)} \setminus \mathcal{N}_{r,s}^{\text{FT}}$	Correction labels for the PPI rectifier.

LLM Fine-Tuning. The base LLM is `Qwen/Qwen2.5-1.5B-Instruct`. Given a review text X, the model forms a pooled representation from the last non-padding token, and a scalar linear head maps this representation to the prediction $f(X)$. Ratings are standardized as

$$y_{\text{scaled}} = (y_{\text{raw}} - 90)/5, \qquad \hat{y}_{\text{raw}} = 90 + 5\hat{y}_{\text{scaled}},$$

so that optimization is stable while all reported residual variances remain interpretable on the original rating scale. We use LoRA adapters on the linear modules and train the scalar head jointly with the adapters.

The standard objective for rating prediction is mean-squared error. In contrast, the downstream PPI estimator is governed by the variance of the residual $Y - f(X)$ after bias correction. We therefore also fine-tune with a residual-variance loss. For a mini-batch of size k, define $r_i = Y_i - f(X_i)$ and $\bar{r} = k^{-1} \sum_{i=1}^{k} r_i$. The variance loss is

$$L_{\text{var}} = \frac{1}{k} \sum_{i=1}^{k} (r_i - \bar{r})^2.$$

The objective is aligned with the PPI correction stage: fine-tuning is used to reduce residual variance, while systematic bias is removed by the correction labels in $\mathcal{N}_{r,s}^{\text{PPI}}$.

6.2 Independent Scaling-Law Validation

The first empirical step is to check the premise behind the allocation rule. Using only the pilot set $\mathcal{H}_{\text{scale}}$, we ask whether the residual variance of LoRA-Var fine-tuning is well approximated by

$$\text{Var}\Big(Y - f^{(s)}(X)\Big) = as^{-\alpha} + b$$

over the range of fine-tuning sizes relevant for the later allocation experiments.

The pilot protocol keeps model selection separate from scaling-law evaluation. In each pilot replication, we randomly partition

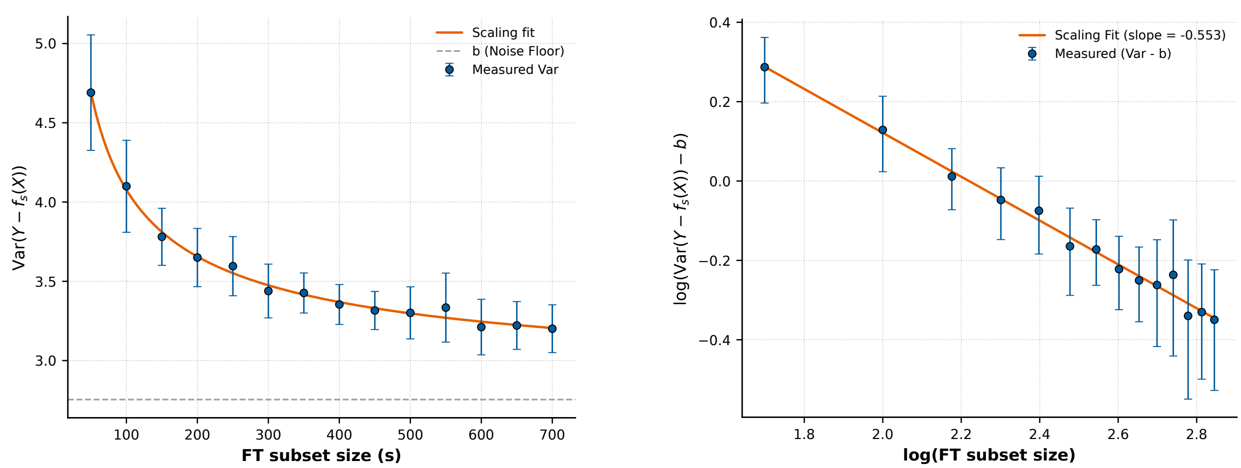
$$\mathcal{H}_{\text{scale}} = \mathcal{V}_{\text{stop}}^{\text{scale}} \,\dot\cup\, \mathcal{V}_{\text{scale}}^{\text{scale}} \,\dot\cup\, \mathcal{T}_{\text{scale}},$$

where $|\mathcal{V}_{\text{stop}}^{\text{scale}}| = 1{,}000$, $|\mathcal{V}_{\text{scale}}^{\text{scale}}| = 1{,}000$, and $|\mathcal{T}_{\text{scale}}| = 8{,}000$. The set $\mathcal{V}_{\text{stop}}^{\text{scale}}$ is used only for early stopping and checkpoint selection. The set $\mathcal{V}_{\text{scale}}^{\text{scale}}$ is used only to measure residual variance and fit the scaling law. The remaining set $\mathcal{T}_{\text{scale}}$ contains the labels from which all fine-tuning subsets are drawn.

We run 10 independent pilot replications. In each replication, the fine-tuning subsets are nested inside $\mathcal{T}_{\text{scale}}$ over

$$s \in \{50, 100, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700\}.$$

For each s, we train a LoRA-Var model on the first s labels in the nested training order and evaluate its residual variance on $\mathcal{V}_{\text{scale}}^{\text{scale}}$. We then fit the power law to the mean residual variance across replications, reported on the original rating scale. To avoid letting a single high-leverage pilot run determine the fitted curve, the reported scaling-law validation uses a symmetric trimmed summary. After training all 10 replications, we remove one upper-tail extreme replication and one lower-tail extreme replication before fitting and plotting the pilot curve. Concretely, the trimmed analysis removes replication 2, which has the largest fitted amplitude and slope, and replication 9, which has the weakest replication-level fit. All numbers in this subsection use the remaining eight replications. Error bars show 95% confidence intervals over these replications.



(a) Power-law fit in linear space

(b) Linear fit in log-log space

Figure 1: Independent validation of the variance-based scaling law on $\mathcal{H}_{\text{scale}}$. The fitted parameters after trimming the two extreme pilot replications are $\hat{a} = 16.86$, $\hat{\alpha} = 0.553$, and $\hat{b} = 2.754$ on the original rating scale.

The main evidence is shown in Figure 1. The fitted curve has $R^2 = 0.995$, and the mean residual variance decreases from 4.69 at $s = 50$ to 3.20 at $s = 700$, far below the constant-predictor baseline of 9.57. The log-log panel gives a second view of the same pattern: after subtracting the fitted noise floor, the residual variance is close to linear in $\log(s)$, with slope $-\hat{\alpha} = -0.553$.

Table 3 checks that this pattern is not only a consequence of averaging across replications. The first rows fit the same power-law form separately within each pilot replication. Brackets report the 25th and 75th percentiles across the trimmed pilot replications. The final row reports an early-to-late holdout check: fitting only the early sizes $s \leq 400$ predicts the later sizes $s \in \{450, \ldots, 700\}$ with median relative error below 3% across replications.

Table 3: Trimmed pilot evidence for the LoRA-Var scaling law

Check	Report
Mean-curve R^2	0.995
Median replication-level R^2	0.955
10th percentile replication-level R^2	0.938
Median fitted exponent $\hat{\alpha}$ [25%, 75%]	0.608 [0.342, 0.700]
Median fitted noise floor $\hat{b}$ [25%, 75%]	2.780 [2.065, 3.055]
Median holdout prediction error for $s > 400$	2.67%

The purpose of the scaling law is allocation, not interpretation of the individual parameters. We therefore also examine the stability of the allocation implied by each replication-level fit. For each pilot replication, we plug its fitted $(\hat{a}_r, \hat{\alpha}_r, \hat{b}_r)$ into the allocation objective used in Section 6.3. Table 4 shows that the resulting fine-tuning fractions remain concentrated around 14–19% of the effective labeled budget, even though the fitted parameters vary across pilot replications.

Table 4: Allocation stability implied by trimmed replication-level scaling-law fits

Budget B	$n_B = 0.8B$	Mean-fit ρ_B^{th}	Median ρ_B^{th}	95% bootstrap interval
300	240	19.2%	18.8%	[15.8%, 22.1%]
500	400	17.0%	16.9%	[14.8%, 19.5%]
700	560	15.7%	15.4%	[13.9%, 17.9%]
1000	800	14.5%	14.2%	[12.6%, 15.6%]

The interval is intentionally conservative because it bootstraps only the eight trimmed pilot replications. Taken together, the pilot results support the premise that additional fine-tuning labels reduce residual variance in a smooth and predictable way. They do not, however, determine the final allocation in the target population. The next two experiments evaluate allocation directly using fresh labeled budgets from $\mathcal{P}_{\text{target}}$. Additional implementation details for the scaling-law fit, including the early-stopping split and reported quantities, are given in Appendix A.

6.3 Full-Grid Allocation Validation

Having established the pilot scaling law, we next test whether it leads to a useful allocation on the target population. The scaling-law parameters are estimated once on $\mathcal{H}_{\text{scale}}$, fixed before the target-budget replications are drawn, and then used to compute a theoretical fine-tuning fraction. We compare that fraction with a full-grid empirical search on fresh labeled budgets from $\mathcal{P}_{\text{target}}$. Replication-specific fitting on target-budget data is left to the ramp-up experiment, where only early trained sizes and the validation split are available for allocation selection. We focus on LoRA-Var + PPI here because the variance-loss scaling law supplies its allocation rule; loss-function comparisons are deferred to Section 6.5.

For each budget B and replication r, we draw a fresh small labeled dataset $\mathcal{N}_r^{(B)}$ from $\mathcal{P}_{\text{target}}$. The validation set is charged to this same budget: we set

$$|\mathcal{V}_r^{(B)}| = 0.2B, \qquad n_B = B - |\mathcal{V}_r^{(B)}| = 0.8B.$$

Only the remaining n_B labels, $\mathcal{N}_{r,\text{eff}}^{(B)} = \mathcal{N}_r^{(B)} \setminus \mathcal{V}_r^{(B)}$, are available for the final FT+PPI estimator. Thus, an allocation fraction ρ corresponds to

$$s(\rho) = \text{round}(\rho n_B) \quad \text{fine-tuning labels, and} \quad n_B - s(\rho) \quad \text{PPI correction labels.}$$

This accounting is important: validation labels are charged to the same labeled budget as all other labels, and they are not returned to the final fine-tuning or correction sets.

Plugging the independent-pilot scaling-law parameters $(\hat{a}, \hat{\alpha}, \hat{b})$ into the FT-PPI allocation objective gives

$$\rho_B^{\text{th}} = \arg\min_{0<\rho<1} \frac{\hat{a}\, s(\rho)^{-\hat{\alpha}} + \hat{b}}{n_B - s(\rho)}, \qquad s(\rho) = \text{round}(\rho n_B).$$

Equivalently, the numerator predicts the residual variance of the LoRA-Var surrogate trained on $s(\rho)$ labels, while the denominator is the number of labels left for PPI correction. We solve this one-dimensional problem numerically for each B after removing the validation labels from the budget. This theoretical rule targets the dominant label-allocation term in equation (2); the empirical full-grid validation below evaluates the full plug-in variance, including the unlabeled prediction-variance term.

To validate this theoretical optimum, we also run a full-grid empirical search over

$$\mathcal{G} = \{0.025, 0.05, 0.075, 0.10, 0.125, 0.15, 0.20, 0.30, 0.40, 0.50, 0.70, 0.90\}.$$

In addition, we train the exact theoretical allocation ρ_B^{th} as an additional candidate. This candidate uses the same replication, the same labeled budget, the same validation set, and the same nested ordering of $\mathcal{N}_{r,\text{eff}}^{(B)}$ as the grid candidates; only the number of fine-tuning labels changes. Thus the LoRA-Var candidate set is

$$\mathcal{G}_B^{\text{Var}} = \mathcal{G} \cup \{\rho_B^{\text{th}}\},$$

which allows the theoretical allocation to compete directly with the original grid points. For each candidate ρ, we fine-tune a LoRA-Var predictor on $\mathcal{N}_{r,s(\rho)}^{\text{FT}}$ and use the remaining $\mathcal{N}_{r,s(\rho)}^{\text{PPI}}$ labels for PPI correction. The main allocation metric is the plug-in estimate of the FT+PPI estimator variance from equation (2):

$$\widehat{V}_{r,B}^{\text{Var}}(\rho) = \frac{\widehat{\text{Var}}_{\mathcal{N}_{r,s(\rho)}^{\text{PPI}}}\{Y - f_{r,\rho}^{\text{Var}}(X)\}}{n_B - s(\rho)} + \frac{\widehat{\text{Var}}_{\mathcal{M}_r^{(B)}}\{f_{r,\rho}^{\text{Var}}(X)\}}{|\mathcal{M}_r^{(B)}|}.$$

Here $\widehat{\text{Var}}_A$ denotes the sample variance over the set A, and $f_{r,\rho}^{\text{Var}}$ denotes the LoRA-Var surrogate trained at allocation ρ in replication r. The first term is the variance contribution from the PPI correction labels, and the second term is the variance contribution from averaging predictions over the unlabeled text pool. This quantity is the empirical counterpart of the allocation objective in the FT-PPI theory.

We then aggregate the objective across the $R = 10$ target-population replications:

$$\overline{V}_B^{\mathrm{Var}}(\rho) = \frac{1}{R} \sum_{r=1}^{R} \widehat{V}_{r,B}^{\mathrm{Var}}(\rho).$$

The full-grid empirical oracle is

$$\rho_B^{\mathrm{oracle}} = \arg \min_{\rho \in \mathcal{G}_B^{\mathrm{Var}}} \overline{V}_B^{\mathrm{Var}}(\rho), \qquad \mathcal{G}_B^{\mathrm{Var}} = \mathcal{G} \cup \{\rho_B^{\mathrm{th}}\}.$$

Including ρ_B^{th} in the candidate set makes the empirical benchmark fair. If the theoretical allocation is the best trained candidate, it is allowed to win the benchmark rather than being compared only with nearby grid points.

Two summaries make the comparison interpretable. Allocation regret measures the loss from using the theoretical allocation rather than the full-grid oracle:

$$\mathrm{Regret}_B = \frac{\overline{V}_B^{\mathrm{Var}}(\rho_B^{\mathrm{th}}) - \overline{V}_B^{\mathrm{Var}}(\rho_B^{\mathrm{oracle}})}{\overline{V}_B^{\mathrm{Var}}(\rho_B^{\mathrm{oracle}})}.$$

The normalized variance ratio compares the theoretical-allocation objective with the Sample Mean objective under the same budget:

$$\mathrm{VarRatio}_B = \frac{\overline{V}_B^{\mathrm{Var}}(\rho_B^{\mathrm{th}})}{\overline{V}_B^{\mathrm{SM}}}.$$

The regret is the primary evidence for allocation accuracy. Exact equality between ρ_B^{th} and ρ_B^{oracle} is not required, because the full-grid objective can be flat over a range of nearby allocations.

Table 5: Full-grid allocation validation using the FT+PPI variance objective

B	n_B	ρ_B^{th}	ρ_B^{oracle}	Regret (%)	Var ratio at ρ_B^{th}
300	240	0.192	0.200	3.69	0.833
500	400	0.170	0.200	0.18	0.720
700	560	0.155	0.125	0.79	0.681
1000	800	0.142	0.100	1.86	0.682

Table 5 reports the theoretical allocation and the full-grid oracle for each labeled budget. Across all budgets, the theoretical rule has low variance regret: below 4% in every case and below 2% for three of the four budgets. The variance ratio at the theoretical allocation is also always below one, indicating that the theoretically chosen LoRA-Var + PPI allocation improves on the Sample Mean objective under the same labeled budget. These results support the allocation rule in the sense needed for inference: even when the allocation fraction is not identical to the full-grid oracle, its objective value is close to oracle.

Figure 2 gives a representative view for $B = 500$. It plots the normalized objective $\overline{V}_B^{\mathrm{Var}}(\rho)/\overline{V}_B^{\mathrm{SM}}$ over the full candidate set. The star marks $\rho_B^{\mathrm{th}} = 0.170$, while the empirical oracle is at $\rho_B^{\mathrm{oracle}} = 0.200$. The two allocations are close in objective value: the regret from using ρ_B^{th} is only 0.18%. Analogous figures for other budgets are reported in Appendix B.

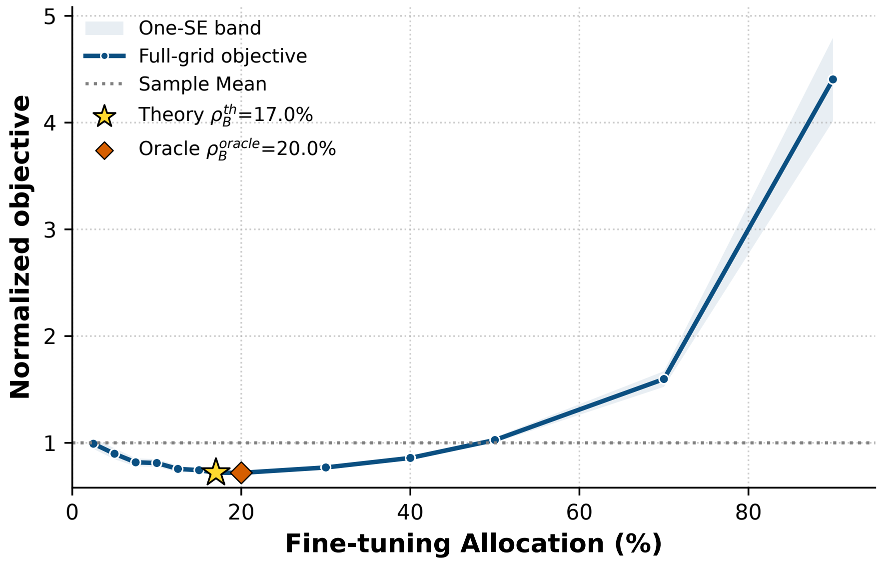


Figure 2: Full-grid LoRA-Var allocation objective for $B = 500$. The vertical axis reports $\overline{V}_B^{\mathrm{Var}}(\rho)/\overline{V}_B^{\mathrm{SM}}$. The star marks the theoretical allocation from the pilot scaling-law fit, and the diamond marks the empirical full-grid oracle.

6.4 Target-Only Ramp-Up Recovery

The full-grid experiment validates the allocation objective, but it is not a deployable procedure because it trains a model at every candidate allocation before deciding which allocation to use. We therefore evaluate a target-only ramp-up procedure for LoRA-Var + PPI. *The full ramp-up procedure should be introduced in the previous "Our Method" section.* The procedure trains only early nested sizes, uses validation residual variance to decide whether further ramp-up is needed, and stops once the fitted objective suggests that the current trained size is large enough.

The ramp-up experiment uses the same target-population label accounting as the full-grid validation. For each budget B, the labeled set $\mathcal{N}_r^{(B)}$ is first split into a validation set $\mathcal{V}_r^{(B)}$ of size $0.2B$ and an effective label set $\mathcal{N}_{r,\mathrm{eff}}^{(B)}$ of size

$$n_B = B - |\mathcal{V}_r^{(B)}| = 0.8B.$$

The validation labels are charged to the labeled budget and are used for allocation selection, but they are not returned to the PPI correction set. This accounting avoids leakage in the final PPI estimator. In practice, these labels are not simply discarded: after the allocation is fixed, the same held-out labels can still be used for checkpoint selection, early stopping, or other training-control decisions for the final fine-tuned model.

Let $s_1 < \cdots < s_L$ denote the nested fine-tuning sizes induced by the allocation grid $\mathcal{G}$ in Section 6.3. At stage ℓ, ramp-up has trained LoRA-Var predictors only at $s_1, \ldots, s_\ell$ and has observed their residual variances on $\mathcal{V}_r^{(B)}$. After at least four stages, it fits

$$\widehat{\sigma}^2(s) = \hat{a}_\ell s^{-\hat{\alpha}_\ell} + \hat{b}_\ell$$

using only the stages observed so far. It then computes the fitted minimizer of

$$\frac{\widehat{\sigma}^2(s)}{n_B - s}.$$

If the fitted minimizer is larger than the current largest trained size s_ℓ, ramp-up proceeds to stage $\ell+1$. If the fitted minimizer is no larger than s_ℓ, ramp-up stops and sets

$$s_B^{\text{ramp}} = s_\ell, \qquad \rho_B^{\text{ramp}} = s_B^{\text{ramp}}/n_B.$$

The final PPI correction set is then $\mathcal{N}_{r,\text{eff}}^{(B)} \setminus \mathcal{N}^{\text{FT}}_{r,s_B^{\text{ramp}}}$. This stopping rule is deliberately simple: every label used in any ramp-up training stage remains inside the final fine-tuning set, so no label that influenced model selection is later treated as an untouched PPI correction label.

We assess ramp-up using the same plug-in objective as in Section 6.3, but only over the original allocation grid $\mathcal{G}$. Let $\rho_B^{\text{oracle},\mathcal{G}}$ be the full-grid oracle over $\mathcal{G}$, and let $\overline{V}_B^{\text{Var}}(\rho)$ be the average plug-in variance across the ten target-population replications. We report

$$\text{Regret}_B^{\text{ramp}} = \frac{\overline{V}_B^{\text{Var}}(\rho_B^{\text{ramp}}) - \overline{V}_B^{\text{Var}}(\rho_B^{\text{oracle},\mathcal{G}})}{\overline{V}_B^{\text{Var}}(\rho_B^{\text{oracle},\mathcal{G}})},$$

$$\text{FitCountSaving}_B = 1 - \frac{\#\{\text{ramp-up trained sizes}\}}{\#\{\text{full-grid trained sizes}\}},$$

$$\text{LabelEffortSaving}_B = 1 - \frac{\sum_{\ell=1}^{\ell_B^{\text{stop}}} s_\ell}{\sum_{\ell=1}^{L} s_\ell}.$$

The first quantity measures allocation quality. The next two quantities measure compute savings at different granularities. Fit-count saving counts how many candidate models are avoided, whereas label-effort saving counts the cumulative number of fine-tuning labels across all trained candidate sizes. The latter is a closer proxy for training effort because a model trained at a larger s is more expensive than a model trained at a smaller s.

Revision note. The more deployable evaluation is to run the ramp-up rule separately within each target-budget replication and then summarize the resulting allocations, regrets, and fit savings across replications. This per-replication replay uses only information available in a single labeled budget, but it is noisier because each scaling-law fit relies on one validation split rather than the aggregate validation curve. I will replace Table 6 with the per-replication version.

Table 6: Target-only ramp-up recovery for LoRA-Var + PPI

B	n_B	Fits used/full grid	ρ_B^{ramp}	$\rho_B^{\text{oracle},\mathcal{G}}$	Regret (%)	Count saving (%)	Label-effort saving (%)
300	240	8/12	0.300	0.200	11.10	33.3	70.9
500	400	8/12	0.300	0.200	7.00	33.3	70.9
700	560	6/12	0.150	0.125	1.60	50.0	85.1
1000	800	5/12	0.125	0.100	2.70	58.3	89.4

Table 6 shows that ramp-up recovers much of the value of the full-grid search while training substantially fewer models. The largest regret occurs at the smallest budget ($B = 300$), where the validation set is small and the early residual-variance curve is noisier. For $B \in \{700, 1000\}$, the regret is below 3% while ramp-up avoids training half or more of the full grid. The label-effort savings are larger than the fit-count savings because the skipped grid points are the largest and most expensive fine-tuning sizes. Thus, the useful target of ramp-up is not exact recovery of the full-grid allocation fraction; it is low objective regret with much lower cumulative fine-tuning effort.

Figure 3 shows the corresponding ramp-up trace for the largest budget ($B = 1000$) in our experiment. The points are the observed validation residual variances at the nested fine-tuning sizes, while the orange curves are the scaling-law fits available after successive ramp-up stages. For $B = 1000$, the procedure stops after fitting the model with $s = 100$ fine-tuning labels; the full-grid oracle over the same deployable grid is $s = 80$. Although the stopped size is not identical to the oracle size, the resulting plug-in variance regret is only 2.70%, illustrating the main role of ramp-up: it stops once the fitted curve indicates that additional large training sizes are unlikely to improve the allocation objective enough to justify their cost. Appendix C provides the analogous trace for $B = 500$.

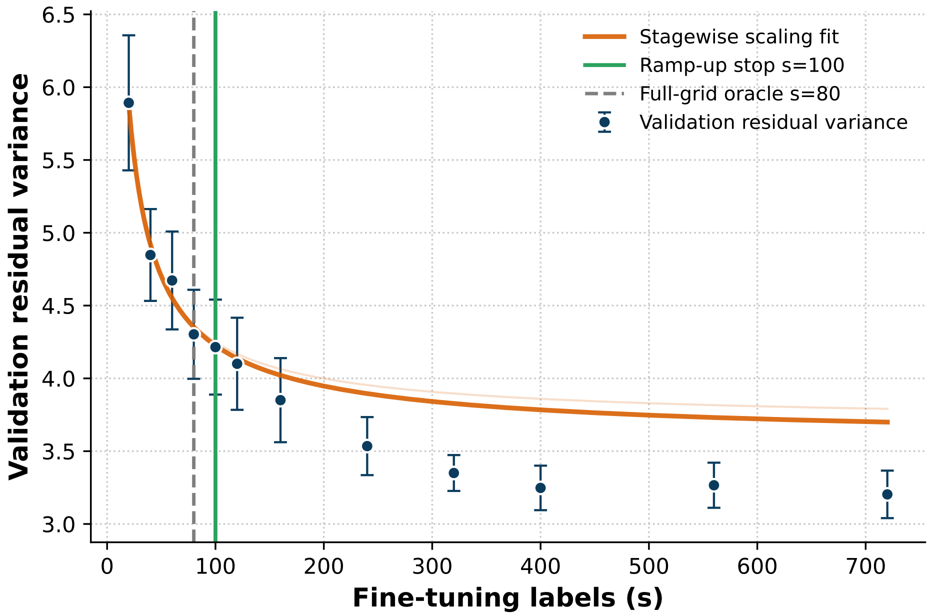


Figure 3: Target-only ramp-up trace for $B = 1000$. The vertical solid line marks the stopped fine-tuning size, and the dashed line marks the full-grid oracle over the same deployable allocation grid.

Grid-design note. The current ramp-up experiment uses the same original allocation grid $\mathcal{G}$ as the full-grid validation. This choice keeps the comparison clean, but it may not be the best deployable design. The ramp-up grid does not have to coincide with the full-grid search points; one could choose a separate set of early training sizes, add intermediate points in the range where the objective is most sensitive, or use a denser low-budget grid. We should run a separate experiment to compare candidate ramp-up grids and determine which set of training sizes gives the best tradeoff between regret and compute.

6.5 End-to-End Estimator Comparison

We now turn from allocation to the final inference task. The central deployable estimator is LoRA-Var + PPI with the target-only ramp-up allocation from Section 6.4. Following the reporting structure of the original empirical analysis, we first compare estimation accuracy and then translate the variance reduction into sample savings. LoRA-MSE + PPI is included as a strong empirical comparator: we train it over the same allocation grid and report the grid point with the smallest plug-in variance objective. We do not propose an allocation rule for LoRA-MSE + PPI, and we do not treat it as a deployable allocation method.

All methods are evaluated on the same target-population replications and budgets

$$B \in \{300, 500, 700, 1000\}.$$

Baselines. We compare against the Sample Mean, which ignores review text and uses all B labeled examples directly. We also include PPI-Only, PPI++-Only, and RePPI as baselines. The two LoRA FT+PPI rows use corrected estimators based on fine-tuned surrogates and differ in how the surrogate is trained and how the fine-tuning size is selected. LoRA-MSE + PPI uses the exhaustive grid comparator, while LoRA-Var + PPI uses the target-only ramp-up allocation. We also report LoRA-Var + PPI at the full-grid best allocation for the plug-in variance objective as a non-deployable lower-bound reference for the allocation search. This row is not an oracle for RMSE or MAE: it is selected by estimated variance, whereas RMSE and MAE are realized point-error summaries across the ten replications.

Estimation accuracy. For each estimator, Table 7 reports RMSE, MAE, and 95% interval coverage, with standard errors in parentheses. Let μ_{target} denote the finite-population mean of $\mathcal{P}_{\text{target}}$ and let $\hat{\mu}_r$ be the estimate from replication r. We compute

$$\text{RMSE} = \sqrt{\frac{1}{R}\sum_{r=1}^{R}(\hat{\mu}_r - \mu_{\text{target}})^2}, \qquad \text{MAE} = \frac{1}{R}\sum_{r=1}^{R}|\hat{\mu}_r - \mu_{\text{target}}|.$$

RMSE standard errors are computed by bootstrap over replications; MAE and coverage standard errors are computed across replications.

Table 7: Estimation accuracy across estimators under different labeled budgets

B	Estimator	Selection	RMSE (SE)	MAE (SE)	Coverage (SE)
300	Sample Mean	–	0.189 (0.038)	0.157 (0.035)	0.90 (0.10)
	PPI-Only	–	–	–	–
	PPI++-Only	–	–	–	–
	RePPI	–	–	–	–
	LoRA-MSE + PPI	var-obj. grid best	0.226 (0.032)	0.196 (0.037)	1.00 (0.00)
	LoRA-Var + PPI	ramp-up	0.141 (0.021)	0.117 (0.026)	1.00 (0.00)
	LoRA-Var + PPI	var-obj. grid best	0.127 (0.023)	0.100 (0.026)	1.00 (0.00)
500	Sample Mean	–	0.142 (0.030)	0.113 (0.029)	0.90 (0.10)
	PPI-Only	–	–	–	–
	PPI++-Only	–	–	–	–
	RePPI	–	–	–	–
	LoRA-MSE + PPI	var-obj. grid best	0.128 (0.027)	0.102 (0.026)	1.00 (0.00)
	LoRA-Var + PPI	ramp-up	0.105 (0.020)	0.083 (0.021)	1.00 (0.00)
	LoRA-Var + PPI	var-obj. grid best	0.121 (0.023)	0.095 (0.025)	1.00 (0.00)
700	Sample Mean	–	0.097 (0.012)	0.088 (0.013)	1.00 (0.00)
	PPI-Only	–	–	–	–
	PPI++-Only	–	–	–	–
	RePPI	–	–	–	–
	LoRA-MSE + PPI	var-obj. grid best	0.139 (0.018)	0.127 (0.019)	0.90 (0.10)
	LoRA-Var + PPI	ramp-up	0.086 (0.014)	0.076 (0.013)	1.00 (0.00)
	LoRA-Var + PPI	var-obj. grid best	0.085 (0.016)	0.072 (0.015)	1.00 (0.00)
1000	Sample Mean	–	0.089 (0.020)	0.065 (0.020)	1.00 (0.00)
	PPI-Only	–	–	–	–
	PPI++-Only	–	–	–	–
	RePPI	–	–	–	–
	LoRA-MSE + PPI	var-obj. grid best	0.068 (0.019)	0.048 (0.016)	1.00 (0.00)
	LoRA-Var + PPI	ramp-up	0.070 (0.020)	0.050 (0.016)	1.00 (0.00)
	LoRA-Var + PPI	var-obj. grid best	0.076 (0.020)	0.054 (0.018)	0.90 (0.10)

Note: PPI-Only, PPI++-Only, and RePPI are reserved baseline rows and are left as placeholders until the corresponding standalone baseline evaluations are finalized. LoRA-MSE + PPI is reported at the grid point with the smallest plug-in variance objective. LoRA-Var + PPI uses the target-only ramp-up allocation, and the LoRA-Var variance-objective grid-best row is a non-deployable reference selected by the plug-in variance objective, not by RMSE or MAE. Coverage is reported for calibration and is not treated as a monotone performance score. Because this pilot table uses only ten matched replications, the RMSE and MAE columns have noticeable Monte Carlo noise; small differences across methods should not be over-interpreted.

Sample savings. Table 8 reports variance reduction and the implied sample savings relative to the Sample Mean. We compute both quantities from estimated variances. If method m has estimated variance $\widehat{\mathrm{Var}}(\hat{\mu}_m)$ and the Sample Mean under the same budget has estimated variance $\widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{SM}})$, then

$$\mathrm{VarReduction}(m) = \mathrm{Savings}(m) = 1 - \frac{\widehat{\mathrm{Var}}(\hat{\mu}_m)}{\widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{SM}})}.$$

The two columns coincide numerically under the usual inverse-B variance scaling assumption. Numbers in parentheses are paired standard errors across the ten matched target-population replications. The final column compares the deployable LoRA-Var ramp-up estimator to the LoRA-MSE grid-best comparator.

Table 8: Variance reduction and sample savings with respect to the Sample Mean

B	Var ramp-up vs. Sample Mean		Var grid best vs. Sample Mean		MSE grid best vs. Sample Mean		Var ramp-up vs. MSE
	Var. red. (%)	Savings (%)	Var. red. (%)	Savings (%)	Var. red. (%)	Savings (%)	Var. red. (%)
300	10.14 (3.28)	10.14 (3.28)	18.95 (3.45)	18.95 (3.45)	-15.43 (5.31)	-15.43 (5.31)	20.70 (4.40)
500	22.87 (2.31)	22.87 (2.31)	27.83 (2.73)	27.83 (2.73)	1.84 (3.24)	1.84 (3.24)	20.73 (3.29)
700	31.39 (1.41)	31.39 (1.41)	32.40 (1.80)	32.40 (1.80)	17.14 (4.10)	17.14 (4.10)	15.51 (4.08)
1000	31.22 (1.24)	31.22 (1.24)	33.02 (0.91)	33.02 (0.91)	20.47 (3.65)	20.47 (3.65)	12.34 (3.18)

The two tables connect the allocation results to the final inference task. Section 6.3 checks whether the LoRA-Var variance objective identifies a good fine-tuning fraction, and Section 6.4 checks whether target-only ramp-up recovers a low-regret approximation to that fraction. Tables 7 and 8 then show how this deployable choice translates into estimation accuracy and label efficiency under the same target-population budgets.

6.6 Combination with CrossPPI

The experiments above study a single-split FT+PPI estimator. CrossPPI can be viewed as a different layer of the same inference pipeline. It is a cross-fitting wrapper: it reuses labeled observations by forming out-of-fold predictions, thereby addressing valid inference with data-dependent predictions and the label waste of a single split PPI estimator. The present experiment instead studies how to improve the LLM surrogate itself: which fine-tuning objective to use, how many labels to spend on fine-tuning, and when to stop ramp-up without training the full allocation grid.

This distinction matters in the LLM setting. CrossPPI treats the learning algorithm as a black box; it does not determine whether the LLM should be fine-tuned with MSE, residual variance, cross-entropy, or another objective. It also does not specify validation accounting for LLM checkpoint selection: the fold whose labels receive out-of-fold predictions cannot also be used to train or select the predictor for that fold. In addition, CrossPPI requires K fine-tuned predictors for K outer folds, which can be substantially more expensive than a single-fit or few-fit deployment. Finally, CrossPPI does not provide a scaling-law budget check or a ramp-up stopping rule that tells the practitioner whether additional fine-tuning labels are still worth spending. These points are not weaknesses of CrossPPI; they clarify that cross-fitting and inference-aware surrogate improvement are complementary rather than substitutes.

A leakage-free combination is straightforward. Partition the target labeled budget into K outer folds. For fold k, do not use labels in fold k to train, tune, or select the predictor that will predict fold k. Instead, fine-tune a LoRA-Var predictor f_{-k} using only the other outer folds, with early stopping based either on an inner validation split drawn from those outer-training folds or on a rule fixed in advance using the independent scaling-law pilot set $\mathcal{H}_{\text{scale}}$. Then form out-of-fold predictions $f_{-k}(X_i)$ for labeled observations in fold k, use the corresponding residuals $Y_i - f_{-k(i)}(X_i)$ for correction, and predict the unlabeled pool by averaging the K predictors. This construction preserves the CrossPPI sample-reuse property while allowing the surrogate in each fold to be trained with the variance-aligned objective.

We therefore view CrossPPI as a compatibility experiment rather than as a replacement for the ramp-up allocation study. A compact empirical check should compare the split FT+PPI estimator from Section 6.5 with CrossPPI versions that use the same LLM architecture and the same target-budget splits, differing only in the cross-fitting wrapper and in the fine-tuning objective. Table 9 gives the reporting structure.

Note: The last two columns should be filled only after running the CrossPPI compatibility experiment, for example at $B \in \{500, 1000\}$. The goal of this check is not to show that one method dominates the other. If CrossPPI-Var gives the shortest confidence intervals, the interpretation is that cross-fitting can further improve efficiency when K LoRA fine-tuning runs are affordable. If split FT+PPI is competitive, the interpretation is that ramp-up provides a compute-light deployment path.

Table 9: Compact report for a CrossPPI compatibility check

Method	Cross-fitting?	FT objective	Selection	# LoRA fits	CI ratio	Coverage
Sample Mean	No	None	None	0	1.000	–
Split FT+PPI	No	Var	ramp-up	J	–	–
CrossPPI-MSE	Yes	MSE	fixed K	K	–	–
CrossPPI-Var	Yes	Var	fixed K	K	–	–

Note: The CI ratio is the average confidence-interval length divided by the Sample Mean confidence-interval length under the same budget. The CrossPPI rows require new cross-fitted LoRA predictions and are placeholders rather than numerical claims.

A Scaling-Law Validation and Implementation Details

Training configuration. All models use `Qwen/Qwen2.5-1.5B-Instruct` with a scalar regression head on the last non-padding token representation. We use LoRA adapters on all linear modules with rank 8, LoRA scaling parameter 16, and dropout 0.05. The LoRA learning rate is 2×10^{-4}, the head learning rate is 5×10^{-4}, weight decay is 0.01, and gradient clipping uses maximum norm 1.0. The maximum sequence length is 192. The experiments were run using UW Hyak computing resources. Training uses at most 20 epochs, with a minimum of 4 epochs and patience 4 for early stopping. The independent scaling-law validation in Section 6.2 and the full-grid allocation validation in Section 6.3 use the same model, LoRA, optimizer, batch-size, and early-stopping hyperparameters; they differ only in the data split, the candidate fine-tuning sizes, and the estimator-level allocation objective being evaluated.

Loss-function residual-variance comparison. In addition to the variance-loss scaling-law run, we evaluate two MSE-based comparisons under the same pilot split protocol: MSE training with MSE-based checkpoint selection, and MSE training with variance-based checkpoint selection. Figure 4 compares the raw-scale validation residual variance across the three training objectives, and Table 10 reports the corresponding scaling-law fits. This comparison uses the same trimmed set of pilot replications as Section 6.2. It shows that all three approaches improve substantially over the constant-predictor baseline, while the variance-loss model with variance-based checkpoint selection gives the lowest residual-variance curve over most fine-tuning sizes.

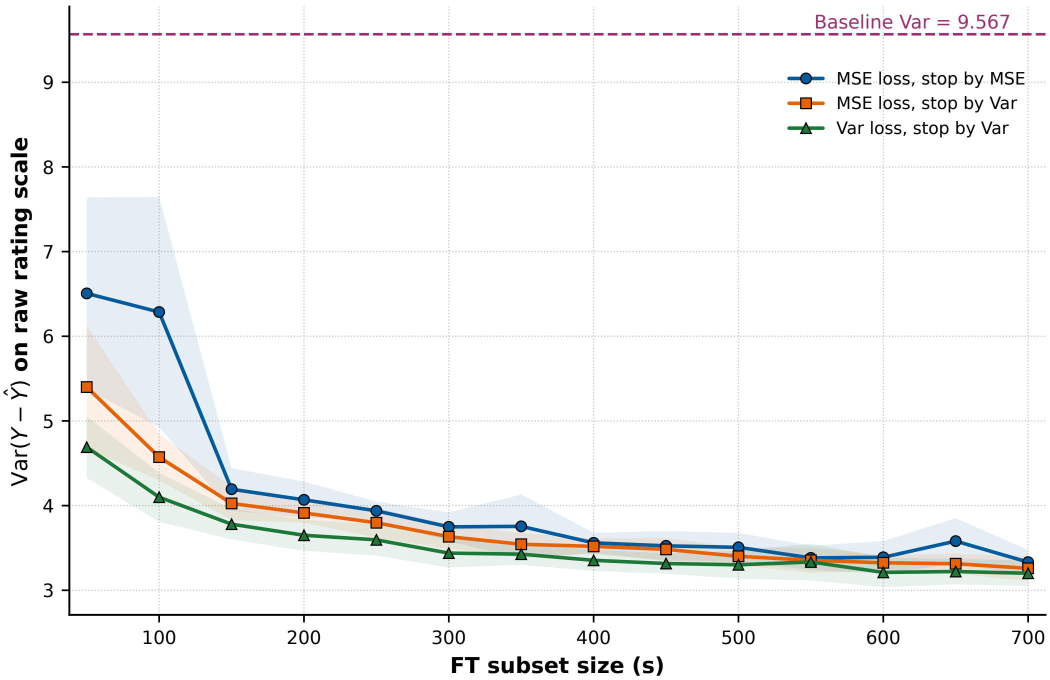


Figure 4: Residual variance on the raw rating scale under MSE and variance fine-tuning objectives. Shaded bands denote 95% confidence intervals across the eight trimmed pilot replications used in Section 6.2.

Table 10: Scaling-law fits for the pilot loss-function residual-variance comparison

Method	$\hat{a}$	$\hat{\alpha}$	$\hat{b}$	R^2
MSE loss, stop by MSE	41.47	0.568	2.266	0.894
MSE loss, stop by Var	26.55	0.583	2.692	0.996
Var loss, stop by Var	16.86	0.553	2.754	0.995

B Additional Full-Grid Allocation Figures

The remaining budget-specific full-grid allocation plots are reported here. Figures 5, 6, and 7 use the same visual convention as Figure 2: the horizontal axis is the fine-tuning allocation fraction after removing validation labels, and the vertical axis is the normalized plug-in variance objective $\overline{V}_B^{\mathrm{Var}}(\rho)/\overline{V}_B^{\mathrm{SM}}$. The star marks the theoretical allocation implied by the Section 6.2 scaling-law fit, and the empirical minimum over the trained candidate set marks the full-grid benchmark.

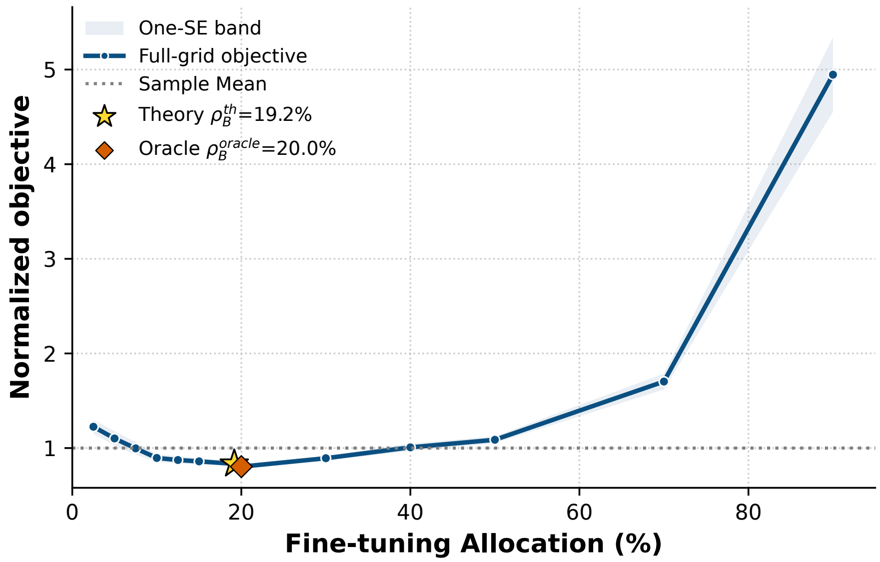


Figure 5: Full-grid LoRA-Var allocation objective for $B = 300$.

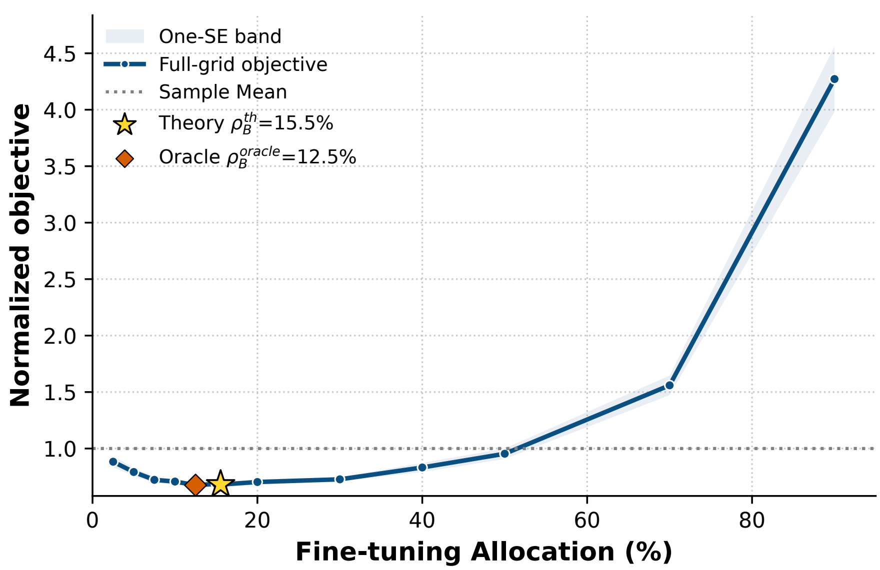


Figure 6: Full-grid LoRA-Var allocation objective for $B = 700$.

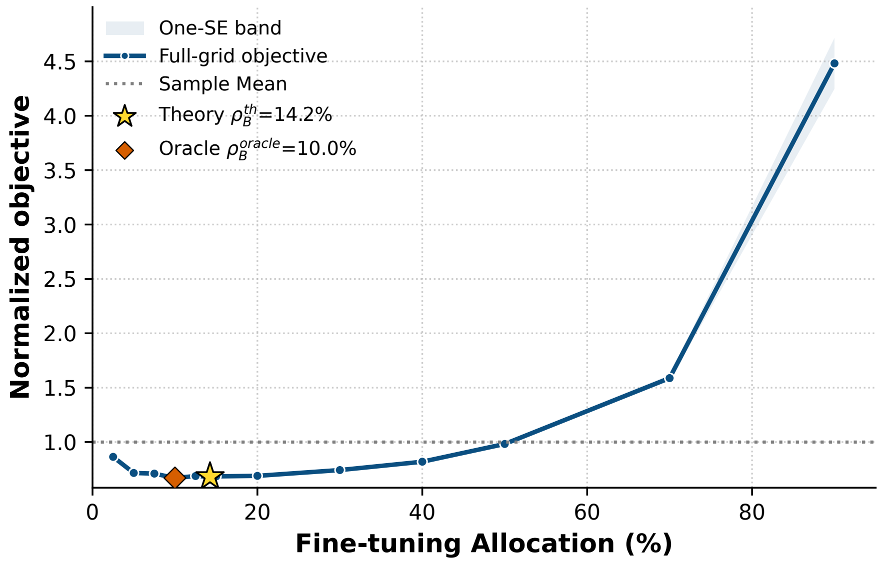


Figure 7: Full-grid LoRA-Var allocation objective for $B = 1000$.

C Target-Only Ramp-Up Trace

Figure 8 provides an additional ramp-up replay using the aggregate validation residual-variance curve across the ten target-population replications. The points show the observed validation residual variances at the nested fine-tuning sizes, and the orange curves show the scaling-law fits after successive ramp-up stages. For $B = 500$, the procedure stops at $s = 120$ while the full-grid oracle over the same deployable grid is $s = 80$; the resulting plug-in variance regret is 7.00%. This trace complements the $B = 1000$ example in Figure 3.

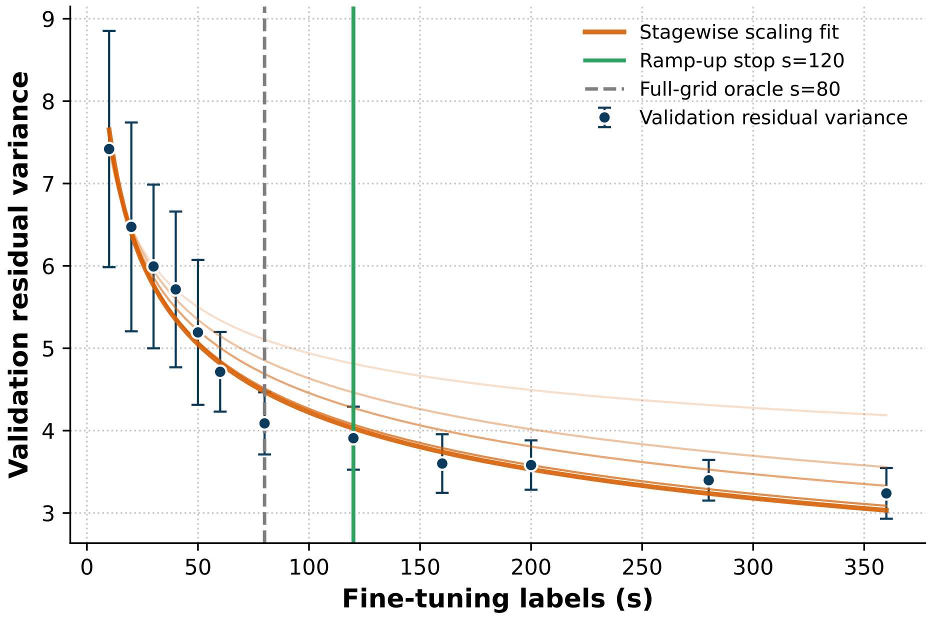


Figure 8: Target-only ramp-up trace for $B = 500$. The vertical solid line marks the stopped fine-tuning size, and the dashed line marks the full-grid oracle over the same deployable allocation grid.